Online appendix to
Special in Repo, Thin on the Ask

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A1. Proofs

The model is in Section 3 of the paper. Equations (1) to (5) and Assumptions 1 and 2 refer to the paper. Write $r_i = v - \gamma y_i$ for dealer i's reservation price for a bond at the GC rate.

Lemma 1. The function $x \mapsto \mathrm{E}[(D_i - x)^+] = \int_x^\infty \Pr(D_i > t)\, dt$ is convex and non-increasing in x. Its derivative is $-\Pr(D_i > x)$ wherever D_i has no atom at x, and its right derivative is $-\Pr(D_i > x)$ everywhere. Differentiating equation (1) gives equation (3). A sell quote lowers the dealer's inventory below y_i. A bid quote at level j raises it at most to $(v + k_i - b_j)/\gamma \leq (v + k - b_2)/\gamma$, by Lemma 2. Under Assumption 1, D_i exceeds every inventory in this range with probability one, so that $\mathrm{E}[(D_i - y)^+] = \mathrm{E}[D_i] - y$ there. This is linear in y, so no condition on the distribution of D_i is needed. Hence $W_i(y) = (v + k_i)\, y - \gamma y^2/2 - k_i\, \mathrm{E}[D_i]$, and $W_i'(y_i) = r_i + k_i$. Because $r_i + k_i = v - \gamma\,(y_i - k_i/\gamma)$, the effect equals that of a fall of k_i/γ in the inventory.

Lemma 2. Let r be the dealer's reservation price. A sell quote of size q at a_j executes with probability π_j. If it executes, the dealer receives $a_j q$ and its inventory falls by q. Because W_i is quadratic with slope r at the current inventory, the change in value is $-rq - \gamma q^2/2$. The expected gain is $\pi_j\,[(a_j - r)\, q - \gamma q^2/2]$. It is largest at $q = (a_j - r)/\gamma$ if $a_j > r$, and at $q = 0$ otherwise, and the largest gain is $\pi_j\,[(a_j - r)^+]^2/(2\gamma)$. A buy quote of size q at b_j gains $\pi_j\,[(r - b_j)\, q - \gamma q^2/2]$, and the same steps give the bid. Under Assumption 2, the two sides of a quote execute in exclusive events. This requires $\pi_j + \pi_{j'} \leq 1$ for the levels j and j' of the ask and the bid, which holds if $\pi_1 \leq 1/2$. The expected gain of the quote is then the sum of the gains of its two sides, so each side is chosen on its own.

Both sides executing. Suppose instead that the ask at level j executes with probability π^a, the bid at level j' with probability π^b, and both with probability $\pi^{ab} > 0$. Suppose also that Assumption 1 covers the larger inventories that this allows. If both sides execute, the inventory changes by $q^b - q^a$, and the quadratic term of W_i adds $\gamma\, q^a q^b$ to the sum of the two gains. The expected gain is

$$\pi^a\,[(a_j - r)\, q^a - \gamma\,(q^a)^2/2] + \pi^b\,[(r - b_{j'})\, q^b - \gamma\,(q^b)^2/2] + \gamma\,\pi^{ab}\, q^a q^b.$$

Let $\rho_a = \pi^{ab}/\pi^a$ and $\rho_b = \pi^{ab}/\pi^b$. Both are at most one, and $\rho_a \rho_b < 1$ makes the gain concave. For positive sizes, the first-order conditions are $q^a = (a_j - r)/\gamma + \rho_a\, q^b$ and $q^b = (r - b_{j'})/\gamma + \rho_b\, q^a$.

Hence

$$\frac{\partial q^a}{\partial r} = -\frac{1-\rho_a}{\gamma\,(1-\rho_a\rho_b)}, \qquad \frac{\partial q^b}{\partial r} = \frac{1-\rho_b}{\gamma\,(1-\rho_a\rho_b)}.$$

Both are at most $1/\gamma$ in absolute value. A rise of k in the reservation price therefore still shrinks the ask and enlarges the bid at given levels, but by less than k/γ. The ask quote is proportional to $N = a_j - r + \rho_a\,(r - b_{j'})$, which falls by $(1-\rho_a)\,k$. Its log therefore falls by $\ln\left(N/(N-(1-\rho_a)\,k)\right)$. With $m = a_j - r \le N$, this is at most $\ln\left(m/(m-k)\right)$. The calibration in Section 5.1 of the paper inverts $\ln\left(m/(m-k)\right)$, so it understates k if both sides can execute.

Lemma 3. For $r < a_1$, both ask gains are positive. The dealer prefers the best ask if and only if $\pi_1(a_1-r)^2 > \pi_2(a_2-r)^2$, that is, $\sqrt{\pi_1}\,(a_1-r) > \sqrt{\pi_2}\,(a_1+\Delta-r)$. Rearranging gives $r < a_1 - \sqrt{\pi_2}\,\Delta/(\sqrt{\pi_1}-\sqrt{\pi_2}) = \bar{r}^a$, because $\pi_1 > \pi_2$. For $a_1 \le r < a_2$, the gain at the best ask is zero, and the dealer quotes behind it. For $r \ge a_2$, it posts no ask. The bid is symmetric. The dealer prefers the best bid if and only if $\sqrt{\pi_1}\,(r-b_1) > \sqrt{\pi_2}\,(r-b_1+\Delta)$, that is, $r > \bar{r}^b$.

Proposition 1. The thresholds are independent of the inventories, so the affected dealers have density $g\,h$ and the others $(1-g)\,h$, with $g = G(S)$. The others do not change. An affected dealer's reservation price rises from r_i to $r_i + k$.

(a) By Lemma 2, an affected ask quote that stays at level j changes from $(a_j - r_i)/\gamma$ to $(a_j - r_i - k)/\gamma$. By Lemma 3, an affected dealer at the best ask stays there if $r_i + k < \bar{r}^a$ and moves behind the best price if $r_i < \bar{r}^a \le r_i + k$. A dealer behind the best ask stays behind it or stops quoting, because its reservation price only rises. Depth at the best ask is the sum of the quotes there. It therefore falls, through fewer quotes and through smaller quotes among those that stay.

(b) The bid is symmetric. An affected bid quote that stays grows by k/γ. An affected dealer behind the best bid moves to it if $r_i \le \bar{r}^b < r_i + k$, and a dealer at the best bid stays there. Depth at the best bid therefore rises, through more quotes and through larger quotes among those that stay.

(c) After the cost, an affected dealer quotes behind the best ask if $\bar{r}^a \le r_i + k < a_2$, with size $(a_2 - r_i - k)/\gamma$. Substituting $u = r_i + k$, the depth of the affected dealers behind the best ask is

$$\frac{g}{\gamma}\int_{\bar{r}^a}^{a_2} (a_2 - u)\,h(u-k)\,du \quad \text{against} \quad \frac{g}{\gamma}\int_{\bar{r}^a}^{a_2} (a_2 - u)\,h(u)\,du \quad \text{before.}$$

For u in $[\bar{r}^a, a_2)$, both $u-k$ and u lie in $[\bar{r}^a - k, a_2)$, where h is non-increasing. Hence $h(u-k) \ge h(u)$, and depth behind the best ask does not fall. On the bid side, the depth of the affected dealers behind the best bid is $(g/\gamma)\int_{b_2}^{\bar{r}^b}(u-b_2)\,h(u-k)\,du$ against $(g/\gamma)\int_{b_2}^{\bar{r}^b}(u-b_2)\,h(u)\,du$. Because h is non-decreasing on $(b_2-k, \bar{r}^b]$, $h(u-k) \le h(u)$, and depth behind the best bid does not rise. With (a), the log of depth at the best ask falls by more than the log of depth behind it. With (b), the log of depth at the best bid rises by more than the log of depth behind it. The integrals count the dealers that stop quoting at a_2 and that start quoting at b_2, so the result holds for any k, not only for a small one.

(d) Total ask depth is the sum of a part that falls and, under (c), a part that does not fall. It can rise or fall. Under (c), its log changes by a weighted average of the changes at the two levels, so it falls by less in proportion than depth at the best ask, or it rises. The bid side is symmetric. The best ask remains a_1 as long as some dealer quotes there, because prices are

discrete. It moves to $a_2 = a_1 + \Delta$ only if every dealer at a_1 moves. The quoted spread then widens by Δ, and the mid-quote $(a+b)/2$ rises by $\Delta/2$. The best bid is symmetric.

Proposition 2. To first order in the size of the order, an aggressive sale at b_1 changes the dealer's payoff by $b_1 - r - c$, and an aggressive purchase at a_1 by $r - a_1 - c$. The dealer sells if and only if the first is positive and buys if and only if the second is positive. Let H be the distribution function of the reservation prices. The share of dealers that sell aggressively is $(1-g)\,H(b_1 - c) + g\,H(b_1 - c - k)$, which is at most $H(b_1 - c)$. The share that buy aggressively is $(1-g)\,[1 - H(a_1 + c)] + g\,[1 - H(a_1 + c - k)]$, which is at least $1 - H(a_1 + c)$.

Proposition 3. An affected dealer's quotes and orders depend on specialness only through $k_i = k\,\mathbf{1}[S > s_i^*]$, so every outcome depends on S only through the set of affected dealers. Once S exceeds every threshold, this set no longer changes. Depth and the number of aggressive orders are sums over dealers. While the best prices do not change, their expected value is $(1-g)$ times the value when no dealer is affected plus g times the value when every dealer is affected. With a continuum of dealers, the best ask can move only once $g = 1$, because the unaffected dealers at the best ask stay there. The best bid does not move, because dealers only join it. The expected change is therefore $g = G(S)$ times the change when every dealer is affected. It is linear in S up to the largest threshold if and only if G is uniform there. A positive quantity that is linear in g has a log that is monotone in g. A ratio of two such quantities, $(c_0 + c_1 g)/(d_0 + d_1 g)$, has the derivative $(c_1 d_0 - c_0 d_1)/(d_0 + d_1 g)^2$, whose sign does not depend on g. Logs and ratios of depth and of the number of quotes are therefore monotone in g, and so in S. So is the average size of the quotes. None of them is proportional to g. The probability of an executed sell on the bond-day is an increasing function of the share of dealers that sell, which is linear in g, so it is also monotone in S. All of these stop changing once S exceeds every threshold.

Units. The comparative statics are in levels of quantity, and the tests use logs. For a dealer that stays at level j, the change in log depth is $\ln\left((a_j - r_i - k)/(a_j - r_i)\right)$ on the ask and $\ln\left((r_i + k - b_j)/(r_i - b_j)\right)$ on the bid. Both are larger in absolute value where the distance between the quote and the reservation price is smaller. In € million the estimates are imprecise, because depth is very skewed across bonds (Appendix A4).

Short-selling rules. Any seller must deliver at settlement. The EU Short Selling Regulation adds a rule for sellers that act neither as market makers nor as authorised primary dealers. Before selling a bond short, such a seller must have borrowed it, agreed to borrow it or have an arrangement with a third party that has located it (Article 13(1) of European Parliament and Council of the European Union 2012). Market makers that have notified their authority are exempt from this rule (Article 17), but a market maker that sells a bond it does not hold must still source it for delivery.

A2. Data notes

Proposal records. The MTS proposal records are stamped in UTC, and I convert them to local time. The book is rebuilt every 15 minutes from 08:15 to 17:30 local time. A populated bond-day has 37.01 snapshots on average. Each proposal carries one bid price, one ask price and their quantities. It also carries an identifier that stays the same while the quote is updated during the day. The records have no dealer identifier, so I cannot follow a dealer across bonds

or days. The rebuild keeps the latest update of each active two-sided proposal. I drop proposals quoted on one side only, which are rare, and snapshots with a zero price or a zero quoted spread. On the days of Appendix A8, a bond has on average 0.49 ask-only and 0.66 bid-only proposals per snapshot, against 15.69 two-sided proposals. The count of active two-sided quotes counts proposals, not dealers. MTS sets a minimum quote size of €2 million for Italian government bonds.

The rebuilt book and the 15-minute best quotes. The 15-minute best quotes of the paper were built separately, with a different program, from the same proposal records. For each bond and 15-minute interval, they average the best prices and the quantities at those prices over the seconds of the interval. Intervals without any change in the best quotes are left out, but the remaining intervals cover 99% of the rebuilt snapshots. Because of the averaging, a rebuilt snapshot can equal the 15-minute values only when the quotes do not move within the interval. Table A1 therefore uses the intervals in which the 15-minute file records a change in a single second. In these intervals, the rebuilt book must also have the same value at the start and at the end of the interval. On these intervals, the two sources agree exactly on the best bid price in 99.75% of cases and on the quantity at it in 98.26%. On the ask side, they agree in 93.36% and 81.19% of cases. The gap on the ask side comes from one-sided proposals. The 15-minute file keeps proposals that quote only an ask and drops those that quote only a bid, while the rebuilt book drops both. On the first 15 days of the sample of Appendix A8, from September 2019 to February 2020, I also rebuilt the book under the rule of the 15-minute file. The two sources then agree exactly on all four values in every one of these intervals. Across bond-days, the daily means of log depth at the best ask from the two sources have a correlation of 0.99, and of 0.96 within bond and day. For bid depth, the correlations are above 0.99 and 0.98. On the same bond-days, best-ask depth falls by 7.78 log points from the GC rate to 2 bp in the 15-minute quotes, which include the ask-only proposals, and by 7.59 in the rebuilt book, which excludes them. The main ask result is thus the same with and without ask-only proposals. On the same 15 days, a one-sided proposal improves on the best two-sided ask in 3.26% of snapshots and on the best two-sided bid in 3.90%.

Fill records. Each record is one execution of a fill-and-kill order against a quote and records the side of the aggressor. The fill records and the order files have 190,238 executions in common. On these, the side agrees in 100.00% of cases. This checks the matching of the two exchange records rather than the side itself. The executed quantities of fully filled orders sum to the order size in 100% of cases. Partially filled orders are 20% of filled orders and execute 67% of their size on average. Against the 15-minute quotes, buy executions are priced above the prevailing mid-quote in 84% of cases and sell executions below it in 85%. I exclude fills of proposals against proposals, which carry no fill-and-kill order. The records identify the side of the aggressive order, not the trader behind it.

Specialness. The GC spot-next rate moves within the day, and SC spot-next trading concentrates in the morning. By 10:00, 64% of the day's SC volume has traded, against 14% of its GC volume (Appendix A7). Morning SC trades can therefore lie above the day's average GC rate even when the bond trades at the GC rate. The paper compares each SC trade with the volume-weighted GC rate of the same 30-minute window. Such a window exists for 52% of SC

trades. Otherwise, the paper uses the nearest 30-minute window of the same day with a GC trade. This window is at most an hour away for 92% of SC trades. The paper then averages over the bond-day with volume weights. With the day's GC rate throughout, 7% of bond-days fall below zero rather than 2%. The two measures have a correlation of 0.99, and 0.99 within bonds. I compare a bond at the GC rate with a bond with 2 bp of specialness on the same bond-days. The whole-day measure gives differences of −14.71 log points for the ratio, −6.59 for ask depth and 8.12 for bid depth. For selling, it gives −7.28 percentage points. The main measure gives −16.39, −7.99, 8.41 and −8.73. With the whole-day measure, the slope of the ratio below zero is −14.20 log points per bp (standard error 6.48). With the main measure, it is −1.36 (standard error 6.67).

Coverage. The regression sample has 36,622 of the 37,080 cash-market bond-days with 15-minute quotes in the pre-COVID and post-PEPP periods, or 98.76%. Of the others, 194 have no SC spot-next trade on the day and 264 have too few past repo days for the controls. The excluded bond-days have thinner books. Within bonds, their best-ask depth is 22 log points lower and their quoted spread 25 log points wider than on included days. They also follow higher specialness, with a median of 4.76 bp on the bond's previous observed day, against 2.03 bp for included days. Because they are only 1.24% of bond-days, they carry little weight in the estimates. These counts start on 17 September 2019, the first day of the regression sample. The comparison of the rebuilt book with the 15-minute quotes in Table A1 uses all 37,986 bond-days with both sources from 2 September 2019 on. Table 7 of the paper uses the 36,746 bond-days with the rebuilt book and the controls.

Clustering. The baseline standard errors are clustered by bond and by day. They allow for persistence within a bond and for common shocks within a day. Special states persist, since 90% of bond-days above 2 bp are followed by another. Bond clustering therefore matters, and day clusters alone give standard errors that are too small.

A3. The ratio and the best ask under eight variations

Table A2 re-estimates the difference between a bond at the GC rate and a bond with 2 bp of specialness for the ask/bid depth ratio at the top of the book and for ask depth at the best quote under eight variations. Columns (1) and (2) change only the standard errors. Column (3) keeps the 95 nominal BTPs. Column (4) replaces the day fixed effects with bond-type-by-day fixed effects. Column (5) drops the three trading days on either side of each quarter-end. Column (6) drops the days from 28 May to 5 June 2020, when two bonds traded above 50 bp of specialness. Columns (7) and (8) move the breakpoint from 2 bp to 1 bp and to 3 bp and report the difference between the GC rate and the breakpoint. The ratio keeps its size and significance in every column. Ask depth at the best quote is significant at the 5% level or better in every column. Among the columns with the breakpoint at 2 bp, the difference in ask depth is smallest for the nominal BTPs alone, at −6.54 log points (standard error 3.03).

Table A1: The rebuilt book and the 15-minute best quotes. Panel A reports the percentage of 15-minute intervals in which the rebuilt book and the 15-minute best quotes agree exactly. The 15-minute file averages the best prices and the quantities at them over the seconds of each interval, so exact agreement is possible only when the quotes do not move. The panel therefore uses the intervals in which the file records a change in a single second and the rebuilt book shows the same value at the start and at the end of the interval. The rebuilt snapshot is the one at the end of the interval. The first row uses all days of the sample and the rebuild of the paper, which keeps only two-sided proposals. The next two rows use the first 15 days of the sample of Appendix A8. They apply the rebuild of the paper and then the rule of the 15-minute file, which keeps proposals that quote only an ask and drops those that quote only a bid. Panel B reports correlations between the bond-day means of 100 times log depth at the best quotes from the two sources, across the 37,986 bond-days of the full-book sample and after removing bond and day means.

	Best bid price (1)	Best ask price (2)	Depth at the best bid (3)	Depth at the best ask (4)
Panel A: Exact agreement on intervals without movement, per cent				
Rebuilt book, all days	99.75	93.36	98.26	81.19
Rebuilt book, 15 days	99.74	90.19	98.84	79.25
Rule of the 15-minute file, same days	100.00	100.00	100.00	100.00
Intervals, all days	5,687	5,555	2,529	2,547
Intervals, 15 days	389	367	172	159
Panel B: Correlation of the bond-day means of log depth				
Across bond-days			1.00	0.99
Within bond and day			0.98	0.96

Table A2: The ratio and ask depth at the best quote under eight variations. This table reports estimates of equation (8) of the paper with bond and day fixed effects. Panel A uses the ask/bid depth ratio at the top of the book and Panel B ask depth at the best quote. The ratio is 100 times the log of ask depth over bid depth. Capped specialness lies between zero and the breakpoint, in bp, and the first row of each panel reports the breakpoint times its coefficient. This is the difference between a bond at the GC rate and a bond at the breakpoint. The breakpoint is 2 bp except in columns (7) and (8), so that the first row is 2β in columns (1) to (6). The parts of specialness below zero and above the breakpoint enter separately, with coefficients per bp. The past dispersion and rate updates are the means over the previous 20 trading days of the log dispersion and of the log number of updates of the bond's SC rate, standardised. Standard errors are clustered by bond and by month in column (1) and by bond and by day in columns (3) to (8). Column (2) uses Driscoll-Kraay standard errors with 5 lags. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Bond × month clusters (1)	Driscoll-Kraay (2)	Nominal BTPs only (3)	Bond type × day FE (4)	No quarter-ends (5)	No 28 May to 5 Jun 2020 (6)	Breakpoint at 1 bp (7)	Breakpoint at 3 bp (8)
Panel A: Ask/bid depth ratio								
GC rate to the breakpoint	−16.39***	−16.39***	−13.72***	−16.45***	−16.46***	−16.34***	−15.11***	−16.18***
	(2.29)	(1.63)	(2.59)	(2.21)	(2.33)	(2.30)	(1.91)	(2.54)
Specialness below 0, per bp	−1.36	−1.36	−0.12	−3.97	−5.45	−1.25	4.00	−4.92
	(5.84)	(5.28)	(4.78)	(7.03)	(8.63)	(6.83)	(6.02)	(7.27)
Specialness above the breakpoint, per bp	0.06	0.06	0.03	0.02	0.03	−0.04	−0.01	0.12*
	(0.06)	(0.05)	(0.08)	(0.09)	(0.08)	(0.20)	(0.10)	(0.06)
Past SC rate dispersion	0.30	0.30	−0.73	−0.01	0.41	0.49	−0.49	0.38
	(1.17)	(0.73)	(1.19)	(1.32)	(1.27)	(1.29)	(1.26)	(1.26)
Past SC rate updates	−3.75**	−3.75***	−1.55	−3.58**	−4.05**	−3.79**	−4.15***	−3.61**
	(1.35)	(0.73)	(1.46)	(1.40)	(1.55)	(1.51)	(1.49)	(1.52)
Panel B: Best-ask depth								
GC rate to the breakpoint	−7.99***	−7.99***	−6.54**	−9.36***	−8.57***	−7.31**	−5.48**	−9.21***
	(2.43)	(1.61)	(3.03)	(2.56)	(2.87)	(2.83)	(2.51)	(3.16)
Bond FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	By type	Yes	Yes	Yes	Yes
Observations	36,622	36,622	27,950	36,622	33,154	35,894	36,622	36,622

A4. The full order book by side and level

Table A3 reports, for each measure of the full order book, the difference between a bond at the GC rate and a bond with 2 bp of specialness in log points, the units of the regressions in the paper. Table 4 of the paper tests the model's predictions on the same measures. Column (2) re-estimates each difference with the measure in € million. These estimates have the signs of the log estimates but are imprecise, because depth is very skewed across bonds. Depth at the best ask has a mean of €45.50 million. From the GC rate to 2 bp, ask depth at the best price falls by €3.04 million, with a standard error of 2.61. Bid depth at the best price rises by €4.71 million, with a standard error of 2.77. Bid depth behind the best price falls by €2.33 million, with a standard error of 1.06. Panel B of Table A11 splits the bonds at the median of their mean best-ask depth, which is €23.37 million in the thinner books and €69.19 million in the others. From the GC rate to 2 bp, best-ask depth falls by 12.08 log points in the thinner books, significant at the 1% level, and by 3.55 in the others, not significant at conventional levels. The difference between the two groups is not significant ($p = 0.18$), and the ask/bid depth ratio falls in both, by 16.84 and 14.40 log points. In euros, best-ask depth falls by €3.55 million in the thinner books and by €2.16 million in the others, and neither estimate is significant. The fall is thus larger in relative terms in thin books, but the data cannot show that it is concentrated there.

Does the depth reappear behind the best ask? From the GC rate to 2 bp, ask depth at the best price falls by 7.83 log points. Depth behind it changes by 1.14 log points (standard error 1.37), and total ask depth by −1.61 (standard error 1.61). On average, 36.82% of ask depth stands at the best price. If all the depth that leaves the best ask reappeared behind it, depth behind the best ask would rise by about 4.56 log points and total ask depth would not change. If none of it reappeared, total ask depth would fall by about 2.88 log points. The estimate for depth behind the best ask is 2.50 standard errors below the first value. This calculation uses the average share at the best price and is therefore only approximate. The estimates in € million are too imprecise to decide.

Depth near the best price. Some proposals stand far from the best price and are unlikely to execute. As in Dufour et al. (2020), who use the quantity at the three best prices, Table A4 limits depth behind the best price to the second and third best prices. These prices hold 62% of the ask depth behind the best price and 62% of the bid depth. From the GC rate to 2 bp, ask depth at the second and third best prices changes by 0.99 log points (standard error 1.80). Depth at the best ask falls by 8.58 log points relative to them, against 8.55 relative to all depth behind the best ask. On the bid side, depth at the best price rises by 12.06 log points relative to the next two prices, against 11.76 relative to all depth behind it. The difference between the best price and the depth behind it is thus as large against the next two prices alone as against all depth behind it.

Dealer inventory. The inventory proxies of the paper are past signed order flow. They cumulate the executed sells minus buys that the quoting dealers absorb. They do so over the previous 5 and 20 days and since the start of the sample, scaled by the bond's past average volume. The proxies behave as inventory should. After the dealers have absorbed net selling over 20 days, bid depth at the best price is lower by 4.04 log points per standard deviation, and

the ask/bid depth ratio is higher by 5.65. I interact capped specialness with the standardised 20-day proxy and control for all three proxies. I then compute the difference in best-ask depth between a bond at the GC rate and a bond with 2 bp of specialness. It is −6.23 log points when the proxy is one standard deviation above its mean. It is −7.62 when the proxy is one standard deviation below its mean. The interaction is 0.69 log points per standard deviation (standard error 1.49, not tabulated). The model does not predict an interaction in quantities. Under Assumption 1 of the paper, the cost raises the reservation price of every affected dealer by k, whatever its inventory, so a quote that stays at its price shrinks by the same quantity. In logs, the same fall is smaller where dealers hold more, because their quotes at the best ask are larger. The interaction has this sign but is not significant at conventional levels. For the ratio, the interaction is 0.39 (standard error 1.48, not tabulated). The proxies use only aggressive orders on MTS and miss the dealers' trades on other venues and with clients. The noise in the proxies also weakens the interaction.

Table A3: Specialness and the full order book, by side and by level. Column (1) reports, for each measure of the full order book, the difference between a bond at the GC rate and a bond with 2 bp of specialness, twice the coefficient β of equation (8) of the paper, in log points or, for the shares, in percentage points. The regressions include bond and day fixed effects, the parts of specialness below zero and above 2 bp, and the past dispersion and rate updates. Column (2) reports the same difference from a regression of the measure in € million. The outcomes are bond-day means of the rebuilt book. The book is rebuilt every 15 minutes from 08:15 to 17:30 local time. Depth behind the best price is total depth minus depth at the best price. The proportional quoted spread is the best ask minus the best bid, divided by the mid-quote. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	GC rate to 2 bp difference (1)	GC rate to 2 bp € million (2)	Observations (3)
Ask/bid depth ratio at the best price	-16.19^{***}		36,622
	(2.34)		
Ask/bid depth ratio behind the best price	5.21^{***}		36,622
	(1.05)		
Ask depth at the best price	-7.83^{***}	−3.04	36,622
	(2.84)	(2.61)	
Bid depth at the best price	8.36^{***}	4.71^{*}	36,622
	(3.09)	(2.77)	
Ask depth behind the best price	1.14	1.03	36,622
	(1.37)	(1.17)	
Bid depth behind the best price	-4.07^{***}	-2.33^{**}	36,622
	(1.29)	(1.06)	
Total ask depth	−1.61	−2.01	36,622
	(1.61)	(3.27)	
Total bid depth	0.87	2.38	36,622
	(1.58)	(3.11)	
Share of the ask side at the best price	-1.46^{**}		36,622
	(0.66)		
Share of the bid side at the best price	2.65^{***}		36,622
	(0.77)		
Active two-sided quotes	0.54		36,622
	(0.37)		
Proportional quoted spread	1.11		36,622
	(2.48)		
Bond and day FE	Yes	Yes	
Spline and past repo controls	Yes	Yes	

Table A4: Depth at the second and third best prices. This table reports the difference between a bond at the GC rate and a bond with 2 bp of specialness, twice the coefficient β of equation (8) of the paper, in log points. The outcomes are bond-day means of the rebuilt book. Depth at the second and third best prices is the quantity at the three best distinct prices minus the quantity at the best price. The regressions use the bond-days of Table 4 of the paper. They include bond and day fixed effects, the parts of specialness below zero and above 2 bp, and the past dispersion and rate updates. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Ask side (1)	Bid side (2)
Depth at the best price	-7.59^{***}	7.72^{**}
	(2.82)	(3.00)
Best price minus all depth behind it	-8.55^{***}	11.76^{***}
	(3.05)	(3.40)
Depth at the second and third best prices	0.99	-4.34^{**}
	(1.80)	(1.89)
Best price minus the second and third	-8.58^{***}	12.06^{***}
	(2.96)	(3.46)
Bond-days	35,737	35,737

A5. Attempted orders from the order files

The order files cover December 2019 to November 2020. They record every fill-and-kill order, filled or not. Table A5 repeats the incidence regressions on attempted orders. Attempted sell orders fall with specialness by about the same amount as executed ones. Their fill rate does not change significantly. Attempted buy orders rise, as executed buys do. The last column repeats the executed-sell regression of the paper on the days that the order files cover.

A6. The repo controls

The paper controls for the bond's repo activity with predetermined measures. These are the means over the previous 20 trading days of the log dispersion and of the log number of updates of the bond's SC rate. The same-day versions can respond to the day's trading. A dealer who sells a bond short, for example, borrows it in spot-next repo on the same day. Table A6 re-estimates the main outcomes on the same bond-days with four sets of controls. The sets are no repo controls, the same-day dispersion, the same-day dispersion and update count, and the predetermined measures of the paper. Four results hold under every set: the fall in ask depth at the best price, the fall in the ask/bid depth ratio, the shift of bid depth to the best price and the fall of selling relative to buying. Without repo controls, the fall in best-ask depth is larger, at 14.86 log points, and executed buying rises. Special bonds have more SC rate updates, and bonds with more updates have less depth at the best ask. Part of the larger estimate without controls thus reflects repo activity in the bond rather than its specialness.

A7. Specialness before 10:00 and the book from 10:00

The paper measures specialness from the whole day's spot-next trades, with each SC trade compared with the GC rate of its half hour. Special spot-next trading starts at about 08:13 local time. By 10:00, 64% of the day's SC volume has traded. Table 8 of the paper instead

Table A5: Attempted orders from the order files, December 2019 to November 2020. This table reports estimates of equation (8) of the paper on the order-file panel, with bond and day fixed effects. The panel has 24,597 bond-days over 233 days. Columns (1) and (4) are indicators, in percentage points, for a bond-day with at least one sell order or at least one buy order, filled or not. Column (2) is a Poisson pseudo-maximum-likelihood regression of the number of sell orders. Its coefficients are semi-elasticities, multiplied by 100 to give log points. Column (3) is the share of sell orders that received a fill, in percentage points, on the bond-days with a sell order. Column (5) is the executed-sell indicator of the paper, from the fill records, on the same days. As in Table 6 of the paper, the first row reports 2β, the difference between a bond at the GC rate and a bond with 2 bp of specialness, and the parts of specialness below zero and above 2 bp have coefficients per bp. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Any sell order (pp) (1)	Sell orders (PPML) (2)	Fill rate of sells (pp) (3)	Any buy order (pp) (4)	Any executed sell (pp) (5)
GC rate to 2 bp of specialness	-7.00^{***}	-24.52^{***}	1.59	4.38^{**}	-6.23^{***}
	(2.08)	(7.06)	(1.03)	(1.86)	(2.13)
Specialness below 0, per bp	−0.28	−25.55	3.67	3.09	−1.27
	(5.69)	(18.94)	(4.18)	(3.47)	(5.61)
Specialness above 2 bp, per bp	0.08	0.24	0.00	0.05	0.10
	(0.11)	(0.23)	(0.02)	(0.12)	(0.10)
Past SC rate dispersion	−1.18	−4.28	−0.59	−1.17	−1.33
	(1.17)	(4.23)	(0.52)	(1.30)	(1.20)
Past SC rate updates	2.57^{*}	6.81	−0.66	4.44^{***}	2.34^{*}
	(1.38)	(4.48)	(0.53)	(1.66)	(1.38)
Bond FE	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes
Observations	24,597	24,597	12,740	24,597	24,597

measures specialness from the SC and GC trades before 10:00. Each SC trade is compared with the GC rate of its half hour or of the nearest half hour before 10:00 with a GC trade. The book and the executed orders are measured from 10:00 on. The regressor is therefore known before the outcomes are observed. Morning specialness exists on 91% of bond-days. Its daily changes within a bond have a correlation of 0.88 with those of the day's specialness. The daily changes of capped specialness under the two measures, the regressor of the paper, have a correlation of only 0.72. The check therefore addresses the order of events within the day. It does not address feedback from order flow to specialness across days, which Appendix A9 addresses with the previous day's order flow. Column (5) also controls for the same outcome before 10:00, so that it asks whether morning specialness predicts the change in the book after 10:00. On the same bond-days, the ratio falls by 14.90 log points with morning specialness and the book from 10:00, against 16.45 for the whole day. With the ratio before 10:00 as a control, it falls by 8.94 log points, significant at the 1% level. Best-ask depth falls by 7.07 log points, against 7.87, and by 5.10 with best-ask depth before 10:00 as a control, significant at the 5% level. The incidence of an executed sell falls by 8.65 percentage points, against 8.69.

A8. Quotes at the best price

Each MTS proposal carries one bid price and one ask price. It also carries an identifier that stays the same while the quote is updated during the day. The number of proposals at the best ask therefore shows whether best-ask depth falls because fewer quotes stand at the best price or because the quotes there are smaller. Table A7 splits log depth at the best price,

Table A6: The difference between a bond at the GC rate and a bond with 2 bp of specialness under four sets of repo controls. Each cell is twice the coefficient β of equation (8) of the paper, with bond and day fixed effects and the parts of specialness below zero and above 2 bp. Column (1) has no repo controls. Column (2) adds the log intraday dispersion of the bond's SC rate on the same day. Column (3) also adds the log number of its SC rate updates on the same day. Column (4) uses the controls of the paper, the means of both measures over the previous 20 trading days. All columns use the same bond-days. Depth is in log points and incidences are in percentage points. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	No repo controls (1)	Same-day dispersion (2)	Same-day dispersion and updates (3)	Predetermined (baseline) (4)
Ask/bid depth ratio, best quotes	-19.26^{***}	-18.02^{***}	-17.26^{***}	-16.39^{***}
	(2.29)	(2.22)	(2.19)	(2.28)
Depth imbalance (pp)	9.17^{***}	8.59^{***}	8.22^{***}	7.49^{***}
	(1.09)	(1.05)	(1.04)	(1.05)
Ask depth at the best quote	-14.86^{***}	-12.11^{***}	-9.94^{***}	-7.99^{***}
	(3.68)	(3.35)	(3.24)	(2.83)
Bid depth at the best quote	4.40	5.90^{*}	7.32^{**}	8.41^{***}
	(3.67)	(3.44)	(3.37)	(3.09)
Quoted spread	4.13	3.11	3.31	1.35
	(2.89)	(2.84)	(2.79)	(2.46)
Ask depth at the best price, full book	-14.50^{***}	-11.76^{***}	-9.64^{***}	-7.83^{***}
	(3.59)	(3.26)	(3.16)	(2.84)
Ask depth behind the best price	-0.58	-0.99	-1.08	1.14
	(1.58)	(1.50)	(1.48)	(1.37)
Bid depth at the best price, full book	4.35	5.80^{*}	7.21^{**}	8.36^{***}
	(3.66)	(3.43)	(3.36)	(3.09)
Bid depth behind the best price	-6.75^{***}	-6.78^{***}	-6.76^{***}	-4.07^{***}
	(1.43)	(1.42)	(1.39)	(1.29)
Bid depth, best minus behind	11.10^{***}	12.58^{***}	13.97^{***}	12.43^{***}
	(4.06)	(3.89)	(3.82)	(3.50)
Any executed sell (pp)	-8.93^{***}	-11.50^{***}	-12.85^{***}	-8.73^{***}
	(1.73)	(1.77)	(1.77)	(1.79)
Any executed buy (pp)	7.31^{***}	2.67^{*}	-0.17	3.64^{**}
	(1.49)	(1.46)	(1.47)	(1.44)
Executed sell minus buy (pp)	-16.24^{***}	-14.17^{***}	-12.68^{***}	-12.37^{***}
	(1.79)	(1.83)	(1.82)	(1.87)
Bond-days			36,622	

snapshot by snapshot, into the log number of quotes at the best price and the log average size of those quotes. It uses every eighth trading day, which gives 42 days and 4,445 bond-days in the regression sample. On average, 5.37 quotes stand at the best ask and 5.41 at the best bid, out of 15.69 active two-sided quotes.

On these days, best-ask depth falls by 6.63 log points from the GC rate to 2 bp. In the 15-minute best quotes, the differences on these days are −6.18 log points for ask depth and 6.59 for bid depth. On all days they are −7.99 and 8.41. The subsample thus gives differences about a fifth smaller than the full sample. Of the fall in best-ask depth, 3.38 log points come from fewer quotes at the best ask and 3.25 from smaller quotes. Neither part is significant on its own. The share of snapshots with a single quote at the best ask rises by 2.56 percentage points from a mean of 14%, significant at the 5% level. At the best bid, depth rises by 9.02 log points. Almost all of this rise comes from more quotes, 8.75 log points, significant at the 1% level. The mean distance of the asks from the best ask changes by 0.77 bp of price (standard error 0.55). The mean distance of the bids from the best bid changes by 0.85 bp (standard error 0.68). Neither change is significant, and the measure is too noisy to show how far the quotes that leave the best ask move. One-sided proposals are rare, and their number does not change significantly with specialness.

Quote moves. Table 5 of the paper follows each proposal from one 15-minute snapshot to the next on the same days. A proposal at the best ask moves away from it if it raises its own ask and stands behind the best ask at the next snapshot. A proposal behind the best ask moves to it if it lowers its own ask and stands at the best ask. A change of status caused only by another dealer's quote is therefore not counted as a move. Proposals that leave the book and new proposals at the best price are counted separately. The bid side is the mirror image. Each rate is the number of moves over the number of proposals at, or behind, the best price, summed over the pairs of snapshots of the bond-day. Bond-days need at least 10 proposals at and 10 behind the best price on the side. Quotes rarely change their size while they stay at the best price: between snapshots, 0.34% of the quotes at the best ask and 0.36% of those at the best bid do so. Proposals do not carry over from one day to the next, so a quote posted on a special day can be smaller from the start. Quote moves therefore cannot show the change in quote size that the model predicts, and the split of depth above measures it only imprecisely.

A9. Direct checks of alternative explanations

Table A8 reports the direct checks of Section 5.7 of the paper. Panel A re-estimates the main differences after two changes. The first change drops the days from the 20th of the month before each Eurex delivery month to the delivery day. The delivery months are March, June, September and December, and the delivery day is the 10th or the next weekday. In these windows, cheapest-to-deliver bonds can become special and basis traders hold them. The windows cover 19% of bond-days. In estimates that are not tabulated, best-ask depth without them is 3.66 log points lower at 2 bp of specialness than at the GC rate with bond-by-week fixed effects, significant at the 10% level. On a day on which capped specialness rises from 0 to 2 bp, best-ask depth changes by −3.65, significant at the 5% level. The second change fixes the repo controls before the bond's current special spell. A spell is a run of consecutive repo days above 0.5 bp of

Table A7: Depth at the best price: the number of quotes and their average size. Column (1) reports twice the coefficient β of equation (8) of the paper. This is the difference between a bond at the GC rate and a bond with 2 bp of specialness. The regressions include bond and day fixed effects, the parts of specialness below zero and above 2 bp, and the past dispersion and rate updates. The first three outcomes are bond-day means over the 15-minute snapshots. They are 100 times the logs of depth at the best price, of the number of proposals at the best price and of their average size. These three logs add up exactly. The other outcomes are the percentage of snapshots with a single proposal at the best price and the mean distance of the two-sided proposals' prices from the best price, in bp of price. The distance is shown for all proposals and for the proposals behind the best price. The last outcome is the number of one-sided proposals per snapshot. Column (2) is the sample mean, geometric for the logs. It is in € million for depth and size and in numbers of quotes for the counts. The sample is every eighth trading day. As in the full-book rebuild, snapshots with a zero price or a zero quoted spread are dropped. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	GC rate to 2 bp (1)	Mean (2)
Best ask		
Depth at the best price, log	-6.63^{*}	23.95
	(3.40)	
Number of quotes at the best price, log	-3.38	4.14
	(2.58)	
Average size of those quotes, log	-3.25	5.79
	(2.31)	
A single quote at the best price (pp)	2.56^{**}	14
	(1.08)	
Mean distance of the asks from the best ask (bp of price)	0.77	4.33
	(0.55)	
The same, asks behind the best price only	0.86	6.41
	(0.71)	
Ask-only proposals (number per snapshot)	-0.63	0.49
	(0.61)	
Best bid		
Depth at the best price, log	9.02^{**}	24.13
	(3.62)	
Number of quotes at the best price, log	8.75^{***}	4.19
	(2.46)	
Average size of those quotes, log	0.28	5.75
	(2.32)	
A single quote at the best price (pp)	-1.77^{*}	14
	(1.05)	
Mean distance of the bids from the best bid (bp of price)	0.85	4.36
	(0.68)	
The same, bids behind the best price only	1.25	6.37
	(0.84)	
Bid-only proposals (number per snapshot)	-1.71	0.66
	(1.75)	
Bond-days	4,445	

specialness. Within a spell, the controls are the trailing 20-day means of the spell's first day, which use only days before the spell. The sample has 2,710 spells, with a median length of 3 repo days and a mean of 15. Panel B adds controls to the same-day change regressions of Table 3 of the paper. The controls are the previous day's change in the outcome, indicators for an executed sell and an executed buy on the previous day, and the previous day's net executed volume.

Recent buying. Demand for a bond could raise its specialness and lower its best-ask depth at the same time. Demand on MTS would show up as executed buying. Panel A of Table A11 interacts capped specialness with an indicator for positive net executed buying over the previous five days. This is the 5-day order-flow measure of Table 4 of the paper with the opposite sign. Net buying is positive on 45% of bond-days. On bond-days without net buying, best-ask depth falls by 8.43 log points from the GC rate to 2 bp, significant at the 1% level. After net buying, it falls by 3.20 log points, not significant at conventional levels. The difference between the two, 5.23 log points, is significant at the 5% level. The ratio and the rebuilt book show the same pattern, and the incidence of an executed sell falls in both groups. With the three order-flow measures of Table 4 as further controls, the fall without net buying is 8.53 log points. The ask result is thus stronger on bond-days without recent buying on MTS.

Auction windows. I collect the Treasury's auction announcements from the Bank of Italy for June 2019 to May 2021. They cover 113 announcements and 231 auctioned bonds, 160 of them reopenings, on 101 ISINs. Of the 126 bonds in the panel, 46 were auctioned during the sample. As in Dufour et al. (2020), an auction window runs from the announcement to the settlement of an auction of the bond. The windows cover 1.73% of bond-days. Within bonds, specialness is 0.67 bp higher on the days between the announcement and the auction than outside the windows. Table A10 re-estimates the main differences without the window bond-days, and inside and outside the windows in one regression. Without the windows, the differences are −16.58 log points for the ratio, −8.07 for ask depth and 8.52 for bid depth, and −8.53 percentage points for selling. Inside the windows the estimates are imprecise, and none of them differs significantly from the estimate outside the windows.

Large entries. A large entry is a move of a bond from at most 0.5 bp to above 3 bp of specialness between consecutive observations at most four days apart. With the day's GC rate, 48 bond-days are large entries, 18 of them on 27 December 2019. On these bond-days, best-ask depth falls by 10.98 log points (standard error 3.89). With the main measure, 26 bond-days are large entries, 11 of them on 30 December 2019. Only 11 of them are among the large entries of the whole-day measure. On these bond-days, best-ask depth changes by 2.00 log points (standard error 4.63). The two measures differ only in the GC rate. Most large entries of the whole-day measure therefore came from moves of the GC rate within the day, not from the bond's own repo rate. The large entries of the main measure are too few and too concentrated on year-ends for two-way clustered standard errors. The paper therefore relies on the continuous same-day change. The change regression compares only two consecutive observations. The stacked event design of Section 5.7 of the paper instead compares the entry day and the two days after it with days −10 to −6, against bonds with stable specialness. There, the same definition gives a change in best-ask depth of −25.43 log points on entry, from the 8 entries that have a control

bond (Appendix A11). Without any December day, the main differences hardly change. They are −8.25 log points for best-ask depth, 7.82 for bid depth and −16.07 for the ratio.

Periods. I estimate one regression with period-specific bond fixed effects and every regressor interacted with a post-PEPP indicator. From the GC rate to 2 bp, the post-PEPP minus pre-COVID difference in best-ask depth is −1.41 log points (standard error 4.46, $p = 0.75$). For the ratio, bid depth and selling, the differences are −1.67, 0.26 and 0.38 ($p = 0.74$, 0.96 and 0.89). Within a bond, capped specialness varies about as much in both periods, with standard deviations of 0.46 and 0.50 bp. The post-PEPP period has more bond-days, 25,725 against 10,897, and more special bonds, with 55% of its bond-days above 2 bp against 40% before COVID.

Specialness below zero. Negative values of the main measure are small. Of these values, 94% lie within 0.5 bp of zero, which is the tick of repo rates. The share within 1 bp of zero is 99%, and the lowest value is −3.65 bp. Negative values also fall on thinly traded bond-days more often. Of these bond-days, 8% have at most five SC trades, against 1% of all bond-days. The slope below zero does not differ significantly from the slope between 0 and 2 bp for any outcome. The differences are 6.84 log points per bp for the ratio ($p = 0.33$), 6.72 for ask depth ($p = 0.40$) and −0.12 for bid depth ($p = 0.98$). For selling, the difference is 3.10 percentage points ($p = 0.61$). Bond-days below zero are 3% of the pre-COVID sample and 2% of the post-PEPP sample.

Valuation. If specialness raised the value of the bond (Duffie, 1996), its price would rise with specialness. I take the bond's closing mid-quote, the last 15-minute snapshot from 16:00 on. I regress the daily change in its log, in bp of price, on the same-day change in capped specialness. The regression has bond fixed effects and bond-type-by-maturity-bucket-by-day fixed effects, which absorb moves of the yield curve. On 35,891 bond-days, a same-day rise of capped specialness from 0 to 2 bp changes the closing mid-quote by −0.60 bp of price, with a standard error of 0.92. With day fixed effects only, the change is −1.11. The next day's change does not respond to today's change in capped specialness, at 0.48 (standard error 0.78). In other limit order markets, the ask/bid depth ratio carries information about future prices (Cao et al., 2009). Here today's ratio predicts the next day's change only weakly, by −0.18 bp per standard deviation (standard error 0.11), not significant at conventional levels. The daily change in the closing mid-quote has a standard deviation of 42.20 bp, and the median price tick is 1 bp of price. The 95% confidence interval of the same-day change runs from −2.40 to 1.21 bp. It rules out upward revaluations larger than 1.21 bp of price for a rise from 0 to 2 bp. It cannot rule out the capitalised cost of borrowing. Two basis points of specialness for a month are worth about 0.17 bp of price.

Tighter fixed effects in the full book. In the full order book, ask depth at the best price relative to depth behind it falls with capped specialness. From the GC rate to 2 bp, it falls by 7.07 log points with bond-by-month fixed effects, significant at the 1% level, and by 4.45 with bond-by-week fixed effects, significant at the 10% level. Bid depth at the best price relative to depth behind it rises by 4.68 with bond-by-month fixed effects, significant at the 5% level. With bond-by-week fixed effects, it changes by 0.24, not significant at conventional levels.

Life cycle and quoting obligations. Table A9 adds time-varying bond characteristics. They are the log of the bond's age, the log of the days since its last issue or reopening, an indicator

for the most recently issued bond of its type and original maturity, the log of the number of reopenings since September 2019 and the number of obliged market makers. With them, the ask/bid depth ratio falls by 15.23 log points from the GC rate to 2 bp and best-ask depth by 10.27, both significant at the 1% level. Best-bid depth rises by 4.95, not significant at conventional levels, and by 5.76, significant at the 5% level, when fixed effects for residual-maturity buckets interacted with days replace the day fixed effects. Within bond-months and with the same controls, the ratio falls by 8.21 log points and best-ask depth by 4.66, both significant at the 1% level, and best-bid depth rises by 3.55, significant at the 5% level. Executed selling falls with the controls, by 7.32 percentage points, but not within bond-months. The response of the ratio is larger where more market makers are obliged to quote the bond. In estimates that are not tabulated, the ratio falls by a further 5.07 log points from the GC rate to 2 bp with one standard deviation more obliged market makers, significant at the 5% level. Most of this comes through best-bid depth, which rises by a further 3.77 log points, significant at the 10% level.

Table A8: Direct checks of alternative explanations. Panel A reports the difference between a bond at the GC rate and a bond with 2 bp of specialness, twice the coefficient β of equation (8) of the paper. The regressions include bond and day fixed effects and the parts of specialness below zero and above 2 bp. The first row drops the futures delivery windows, from the 20th of the month before each Eurex delivery month to the delivery day, and uses the past dispersion and rate updates. The second row fixes these two controls at their values on the first day of the bond's current special spell, a run of consecutive repo days above 0.5 bp of specialness. Panel B reports twice the coefficient on the same-day change in capped specialness, the change in the outcome on a day on which capped specialness rises from 0 to 2 bp. The regressions explain the same-day change in the outcome, with bond and day fixed effects and the past dispersion and rate updates. The first row repeats the same-day change regression of Table 3 of the paper on the bond-days on which the previous day's change and order flow exist. The second row adds the previous day's change in the outcome, indicators for an executed sell and an executed buy on the previous day, and the previous day's net executed volume. Depth is in log points and incidences are in percentage points. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Ask/bid depth ratio (1)	Best-ask depth (2)	Best-bid depth (3)	Any executed sell (pp) (4)
Panel A: GC rate to 2 bp, bond and day fixed effects				
Without BTP futures delivery windows	-17.20^{***}	-9.00^{***}	8.20^{***}	-8.97^{***}
	(2.63)	(2.80)	(3.01)	(1.85)
Repo controls fixed before the current special spell	-15.44^{***}	-9.86^{***}	5.58	-7.52^{***}
	(2.61)	(3.47)	(3.67)	(1.82)
Panel B: Same-day changes, rise in capped specialness from 0 to 2 bp				
Controls of Table 3 of the paper, same bond-days as the next row	-5.33^{***}	-3.17^{**}	2.16	2.75
	(1.71)	(1.60)	(1.81)	(2.71)
Adding the previous day's change and order flow	-4.49^{***}	-3.79^{***}	0.62	2.00
	(1.29)	(1.29)	(1.53)	(1.71)

A10. Supply and demand in the repo market

Specialness can rise because more traders want to borrow a bond or because fewer holders lend it. As Cohen et al. (2007) do for equity lending, I separate the two with the quantity of SC repo traded. A change in specialness is demand-type if specialness and the log SC spot-next volume of the bond move in the same direction, and supply-type if they move in opposite directions. The MTS Repo records also give the initiating side of each trade. A trade initiated by the borrower of the bond executes at a lower repo rate than one initiated by its lender, by 0.58 bp on average

Table A9: Bond life cycle and quoting obligations. This table reports the difference between a bond at the GC rate and a bond with 2 bp of specialness, twice the coefficient β of equation (8) of the paper, with further controls and fixed effects. Column (1) repeats Panel A of Table 3 of the paper. Column (2) adds the log of the bond's age, the log of the days since its last issue or reopening and an indicator for the most recently issued bond of its type and original maturity. It also adds the log of the number of reopenings since September 2019 and the number of market makers obliged to quote the bond that month. The bond characteristics come from the monthly bond reference files of MTS and the reopenings from the Treasury's auction announcements. Column (3) replaces the day fixed effects with fixed effects for residual-maturity buckets interacted with days. The buckets are 0 to 2, 2 to 3.5, 3.5 to 5.5, 5.5 to 8, 8 to 12, 12 to 20 and more than 20 years. Column (4) uses bond-by-month and day fixed effects, as in Panel B of Table 3 of the paper, and column (5) adds the controls of column (2) except the number of obliged market makers, which changes only from month to month. Columns (2) and (3) use the bond-days with obligation data. Depth and the quoted spread are in log points and the executed sell in percentage points. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Baseline	Life cycle and obligations	Life cycle and obligations	Bond × month FE	Bond × month, life cycle
	(1)	(2)	(3)	(4)	(5)
Ask/bid depth ratio	-16.39^{***}	-15.23^{***}	-16.18^{***}	-8.86^{***}	-8.21^{***}
	(2.28)	(2.34)	(2.05)	(1.70)	(1.68)
Best-ask depth	-7.99^{***}	-10.27^{***}	-10.42^{***}	-5.39^{***}	-4.66^{***}
	(2.83)	(2.91)	(2.17)	(1.53)	(1.49)
Best-bid depth	8.41^{***}	4.95	5.76^{**}	3.47^{**}	3.55^{**}
	(3.09)	(3.07)	(2.33)	(1.65)	(1.70)
Quoted spread	1.35	1.48	−0.64	0.70	0.88
	(2.46)	(2.54)	(2.24)	(1.73)	(1.75)
Any executed sell (pp)	-8.73^{***}	-7.32^{***}	-6.46^{***}	−0.22	−0.02
	(1.79)	(1.77)	(1.71)	(1.91)	(1.92)
Life-cycle controls	No	Yes	Yes	No	Yes
Obliged market makers	No	Yes	Yes	No	No
Bond FE	Yes	Yes	Yes	No	No
Bond × month FE	No	No	No	Yes	Yes
Day FE	Yes	Yes	No	Yes	Yes
Maturity bucket × day FE	No	No	Yes	No	No
Bond-days	36,622	36,316	36,316	36,622	36,622

Table A10: Auction windows. This table reports twice the coefficient β of equation (8) of the paper, the difference between a bond at the GC rate and a bond with 2 bp of specialness. The regressions include bond and day fixed effects, the parts of specialness below zero and above 2 bp, and the past dispersion and rate updates. An auction window of a bond runs from the announcement to the settlement of an auction of that bond, from the Bank of Italy's announcements of Treasury auctions. The first row uses all bond-days and the second drops the window bond-days. The last two rows come from one regression on all bond-days that adds a window indicator and its product with capped specialness. They report the difference inside the windows and the difference between inside and outside. Depth is in log points and selling in percentage points. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Ask/bid depth ratio	Best-ask depth	Best-bid depth	Any executed sell (pp)
	(1)	(2)	(3)	(4)
All bond-days (baseline)	-16.39^{***}	-7.99^{***}	8.41^{***}	-8.73^{***}
	(2.28)	(2.83)	(3.09)	(1.79)
Without auction windows	-16.58^{***}	-8.07^{***}	8.52^{***}	-8.53^{***}
	(2.31)	(2.80)	(3.15)	(1.80)
Inside auction windows	−8.85	−2.29	6.56	−8.72
	(6.86)	(8.42)	(7.30)	(5.53)
Inside minus outside	7.75	5.85	−1.90	0.01
	(6.69)	(8.22)	(7.63)	(5.52)
Bond-days without windows	35,989	35,989	35,989	35,989

Table A11: Recent buying and thin books. Each cell is the difference between a bond at the GC rate and a bond with 2 bp of specialness, twice the coefficient β of equation (8) of the paper. Panel A adds an indicator for positive net executed buying over the previous five days and its product with capped specialness. Panel B adds the product of capped specialness with an indicator for the bonds whose mean best-ask depth in the 15-minute quotes lies below the median across bonds. In each panel, the first two rows report the difference for each group and the third row the difference between the groups. Columns (1) and (2) use the 15-minute best quotes. Columns (3) and (4) use the rebuilt book, in log points and in € million. Column (5) uses the executed orders, in percentage points. The regressions use the bond-days of Table 4 of the paper, with bond and day fixed effects and the controls of equation (8) of the paper. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Best-ask depth (1)	Ask/bid depth ratio (2)	Best-ask depth, full book (3)	Best-ask € million (4)	Any executed sell (pp) (5)
Panel A: Net executed buying over the previous five days					
No net buying	-8.43^{***}	-16.41^{***}	-8.05^{***}	-3.64	-8.42^{***}
	(2.85)	(2.48)	(2.86)	(2.87)	(1.94)
Net buying	-3.20	-9.55^{***}	-3.47	-0.13	-7.10^{***}
	(3.32)	(2.87)	(3.36)	(2.65)	(1.98)
Difference	5.23^{**}	6.85^{**}	4.59^{*}	3.52^{*}	1.32
	(2.57)	(3.21)	(2.57)	(1.93)	(1.70)
Bond-days	35,737	35,737	35,737	35,737	35,737
Panel B: The bond's mean best-ask depth					
Above the median	-3.55	-14.40^{***}	-3.45	-2.16	-9.49^{***}
	(4.00)	(2.97)	(4.01)	(3.90)	(2.30)
Below the median	-12.08^{***}	-16.84^{***}	-11.80^{***}	-3.55	-7.62^{***}
	(4.50)	(3.18)	(4.44)	(2.98)	(2.26)
Difference	-8.52	-2.43	-8.34	-1.40	1.87
	(6.27)	(4.12)	(6.19)	(4.53)	(2.87)
Bond-days	35,737	35,737	35,737	35,737	35,737

within a bond-day and on 79% of bond-days, as a borrower that accepts a lender's quote should. Table A12 regresses the change in each outcome on the change in capped specialness for each type, with the type indicator, the predetermined repo controls and bond and time fixed effects. It reports the change in the outcome when capped specialness rises from 0 to 2 bp, the scale of Table 3 of the paper. Panel A uses daily changes and the day's SC volume. Panel B classifies the change by morning specialness and morning SC volume, from the trades before 10:00, when 64% of SC volume has traded, and measures the outcomes from 10:00. The day's cash trading after 10:00 therefore cannot drive the classification. Panels C and D use weekly and monthly means.

Best-ask depth falls mainly with demand-type changes. At daily and monthly frequency, it changes by -7.42 and -20.01 log points with demand-type changes, against 0.84 and -2.57 with supply-type changes. Within weeks, best-ask depth falls with both types, by -4.79 log points for supply-type changes, significant at the 10% level, and by -12.11 for demand-type changes, significant at the 1% level. Classified before 10:00, neither type moves best-ask depth significantly. Best-bid depth rises with supply-type changes at daily and monthly frequency, by 6.04 and 23.66, but not within weeks or with the morning classification. The ask/bid depth ratio falls with both types at daily, weekly and monthly frequency. Part of the daily classification may reflect the same day's cash trading, because a short sale is covered by borrowing the bond in spot-next repo on the same day. Executed selling rises with demand-type changes in Panel A, by 9.63 percentage points, but not in Panel B. Table A12 also shows that the quoted spread

widens with supply-type changes in Panel A and narrows with demand-type changes in Panel C, and that executed selling rises with supply-type changes in Panel B.

A11. The event study

Table A13 reports the stacked event study of Section 5.7 of the paper. An entry is a bond-day with specialness above 2 bp after five consecutive observations of the bond at or below 1 bp within 14 calendar days, with specialness above 1 bp on at least two of the next three observations. The sample has 26 entries on 22 bonds. Each entering bond is compared with bonds whose specialness moves by at most 1 bp over the same 21 trading days, which 22 of the entries have. On the entry day and the two days after it, specialness is 1.58 bp higher and best-ask depth 24.92 log points lower than on days −10 to −6. In the five days before the entry, best-ask depth is 7.25 log points lower, which is not significant, but the estimate is too imprecise to rule out a fall of the size of the baseline estimate. Best-bid depth is 14.49 log points lower (standard error 13.96) on entry, against the rise that the model predicts, and the ratio does not change significantly. In the following week, executed selling rises and the quoted spread narrows. The fall in best-ask depth also fades while specialness stays about 1 bp higher.

Table A14 repeats the event study under other entry rules, with the same stacked design. The rules vary the level of specialness before the entry, at most 0.5 or 1 bp, the number of observations at that level, three or five, the jump, above 1.5, 2 or 3 bp, and whether specialness must stay above 1 bp on two of the next three observations. Of the 18 rules with at least eight entries, one, with 8 entries of which 3 have a control bond, gives no estimate on entry, and the table shows the other 17. Entries without a control bond, a bond whose specialness moves by at most 1 bp over the window, do not contribute to the estimates, so the table also reports the number of entries with a control. Best-ask depth falls on entry under all 17 rules. The fall is significant at the 10% level under 7 of them with two-way clustered standard errors and under 10 with a wild-cluster restricted bootstrap by bond. The rules are nested, so they are not independent checks, and two of them give identical estimates because they share the same six entries with a control, so that their bootstrap p-values differ only by simulation noise. With a jump above 1.5 bp after at most 1 bp, the rules give 77 to 229 entries, and best-ask depth also falls in the five days before the entry, with bootstrap p-values below 0.03. Across all rules, this fall before the entry is significant at the 10% level with the bootstrap under 8 of the 17 rules. Specialness in those days is, if anything, lower, partly because the rule bounds specialness before the entry but not in the reference days. The last row uses the large-entry definition of Appendix A9, a move from at most 0.5 bp to above 3 bp between consecutive observations. Two of the 26 large entries of Appendix A9 fall on 20 and 26 September 2019. These dates lie in the first ten trading days of the sample, too early for the ten days before the entry that the stacked design needs. In this design, only 8 of the other 24 entries have a control bond, and they give a change of −25.43 log points on entry.

Table A15 lists the entries of the main rule. The number of entries in auction windows is 1, in futures delivery windows 2 and at the turn of the year 3. Without the entries in auction windows and at the turn of the year, best-ask depth changes by −24.33 log points on entry, with a standard error of 12.78. Controls matched on bond type and residual-maturity bucket leave

Table A12: Supply-type and demand-type changes in specialness. A change in specialness is demand-type if specialness and the log SC spot-next volume of the bond move in the same direction, and supply-type if they move in opposite directions. Each cell reports twice the coefficient on the change in capped specialness for that type, the change in the outcome when capped specialness rises from 0 to 2 bp, or the difference between the two types, from a regression of the change in the outcome with the type indicator, the predetermined repo controls and bond and time fixed effects. Panel A uses changes between consecutive observations of a bond at most four calendar days apart and the day's SC volume. Panel B classifies by morning specialness and morning SC volume, from the trades before 10:00, and uses outcomes from 10:00 on. Panels C and D use weekly means, for bond-weeks with at least three bond-days, and monthly means, for bond-months with at least eight bond-days, of consecutive weeks or months. The time fixed effects are day, week and month fixed effects. Depth and the quoted spread are in log points and the executed sell in percentage points. Standard errors clustered by bond and by time period are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Supply-type (1)	Demand-type (2)	Difference (3)
Panel A: Daily changes, by the day's SC volume			
Ask/bid depth ratio	−5.20**	−5.83**	0.63
	(2.18)	(2.25)	(2.76)
Best-ask depth	0.84	−7.42***	8.26***
	(2.27)	(1.95)	(2.83)
Best-bid depth	6.04***	−1.59	7.63***
	(2.25)	(2.05)	(2.30)
Quoted spread	11.51***	1.82	9.69**
	(2.84)	(3.19)	(3.74)
Any executed sell (pp)	−2.79	9.63***	−12.41***
	(3.39)	(3.67)	(4.48)
Observations		35,911	
Panel B: Daily changes, classified before 10:00			
Ask/bid depth ratio	−4.50*	−1.13	−3.38
	(2.56)	(2.56)	(2.99)
Best-ask depth	−1.84	−1.20	−0.64
	(2.14)	(2.03)	(2.86)
Best-bid depth	2.66	−0.07	2.73
	(2.08)	(1.48)	(1.93)
Any executed sell (pp)	6.70**	0.45	6.25
	(3.21)	(3.24)	(4.10)
Observations		31,821	
Panel C: Weekly changes			
Ask/bid depth ratio	−7.19**	−13.13***	5.94
	(2.79)	(4.20)	(5.00)
Best-ask depth	−4.79*	−12.11***	7.32*
	(2.77)	(3.45)	(4.21)
Best-bid depth	2.40	1.02	1.38
	(3.07)	(3.36)	(4.33)
Quoted spread	5.49	−9.80**	15.29***
	(4.31)	(3.70)	(5.40)
Any executed sell (pp)	−3.21	−4.03	0.82
	(3.93)	(4.16)	(4.51)
Observations		6,968	
Panel D: Monthly changes			
Ask/bid depth ratio	−26.23***	−25.69***	−0.54
	(5.11)	(5.65)	(8.51)
Best-ask depth	−2.57	−20.01***	17.44**
	(5.64)	(5.19)	(7.59)
Best-bid depth	23.66***	5.68	17.98*
	(6.38)	(4.75)	(8.89)
Quoted spread	0.13	−0.76	0.90
	(4.48)	(5.50)	(6.09)
Any executed sell (pp)	−15.91***	−10.88***	−5.04
	(5.40)	(3.21)	(6.71)
Observations		1,777	

only 2 stacks with a control, too few to estimate. With 55 bonds in the stacks, two-way clustered standard errors may be unreliable. I therefore also use a wild-cluster restricted bootstrap by bond, with Webb weights and 1,999 draws. It gives $p < 0.01$ for the fall in best-ask depth on entry and $p = 0.23$ for the days before the entry. For the difference between the two, $p = 0.05$, and for the ratio on entry, $p = 0.40$.

Table A13: Entries into specialness. This table reports a stacked event study of entries into specialness. An entry is a bond-day with specialness above 2 bp after five consecutive observations of the bond at or below 1 bp within 14 calendar days, with specialness above 1 bp on at least two of the next three observations. Each entry forms a stack of the 21 trading days from 10 days before to 10 days after the entry day. The stack contains the entering bond and control bonds whose specialness moves by at most 1 bp over these days and that have no entry nearby. Each column reports the coefficient on the entering bond in the days indicated, relative to days −10 to −6, from a regression with stack-by-bond and stack-by-day fixed effects. SC repo volume is 100 times the log of the bond's SC spot-next volume, and the borrower-initiated share is the share of that volume in trades initiated by the borrower of the bond. Depth and the quoted spread are in log points, the executed orders in percentage points and net executed buying in € million. The sample has 26 entries on 22 bonds and 2,756 stacked bond-days. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Days −5 to −1 (1)	Days 0 to +2 (2)	Days +3 to +10 (3)
Specialness (bp)	−0.12	1.58***	1.06***
	(0.11)	(0.16)	(0.19)
SC repo volume	1.18	16.24	13.93
	(10.29)	(16.67)	(15.02)
Borrower-initiated share (pp)	0.21	−13.87**	−12.88***
	(3.99)	(5.27)	(4.42)
Ask/bid depth ratio	−1.06	−10.44	4.06
	(8.36)	(13.97)	(9.41)
Best-ask depth	−7.25	−24.92**	−8.08
	(6.88)	(11.32)	(7.64)
Best-bid depth	−6.19	−14.49	−12.14
	(8.26)	(13.96)	(9.89)
Quoted spread	−11.32	−11.09	−24.18**
	(9.59)	(8.60)	(10.57)
Any executed sell (pp)	2.57	1.14	16.88*
	(9.85)	(10.65)	(8.58)
Any executed buy (pp)	2.46	3.82	0.98
	(6.99)	(10.47)	(7.69)
Net executed buying (€ million)	14.28	12.69	0.58
	(16.15)	(15.09)	(10.25)

A12. Special and borrowed

Section 5.7 of the paper asks whether repo borrowing alone reduces best-ask depth. In Panel A of Table A16, a bond-day has above-average borrowing if the log of the bond's borrower-initiated SC spot-next volume exceeds its mean over the previous 20 trading days, and below-average borrowing otherwise. Panel B instead compares the mean over the previous five trading days with the mean over the 20 trading days before them, so that the classification does not use the day's trading. Columns (1) to (3) compare special bonds, above 2 bp, and bonds at the GC rate, at most 0.5 bp, in each borrowing state with bonds at the GC rate that have below-average borrowing. The regressions also include an indicator for the bond-days between 0.5 and 2 bp.

Table A14: Entry rules for the event study. Each row repeats the stacked event study of Table A13 with another entry rule. An entry is a bond-day with specialness above the jump after the given number of consecutive observations at or below the given level, within 14 calendar days for five observations, and, where stated, with specialness above 1 bp on at least two of the next three observations. The column headings give the days relative to the entry day. Columns (3) to (5) report the coefficients for best-ask depth on days −5 to −1, 0 to +2 and +3 to +10, relative to days −10 to −6. Column (6) reports the ratio on days 0 to +2 and column (7) specialness on days −5 to −1, in bp. Columns (8) and (9) report p-values for best-ask depth on days −5 to −1 and 0 to +2 from a wild-cluster restricted bootstrap by bond, with Webb weights and 1,999 draws. The regressions have stack-by-bond and stack-by-day fixed effects. Column (2) reports the number of entries and, in parentheses, the number with at least one control bond. Entries without a control bond do not contribute to the estimates. Rules with fewer than eight entries are omitted, and so is one rule whose entries give no estimate. The last row uses the large-entry definition of Appendix A9. The stars use standard errors clustered by bond and by day. *, ** and *** denote significance at the 10%, 5% and 1% levels.

		Best-ask depth			Ratio	Specialness	Bootstrap p	
Entry rule	Entries	−5 to −1	0 to +2	+3 to +10	0 to +2	−5 to −1	−5 to −1	0 to +2
≤0.5 bp for 3, >1.5 bp, persistent	16 (11)	−5.80	−19.22	−16.08	−15.45	−0.06	0.62	0.37
≤0.5 bp for 3, >1.5 bp	40 (30)	−17.46	−30.21*	−15.45	−11.05	−1.96	0.03	<0.01
≤0.5 bp for 3, >2 bp	16 (8)	−14.09	−28.93	−1.54	17.36	−0.10	0.13	0.10
≤0.5 bp for 5, >1.5 bp	20 (17)	−29.07*	−45.83**	−24.02	−15.85	−0.20	<0.01	<0.01
≤0.5 bp for 5, >2 bp	8 (5)	−22.58	−33.17	−5.79	9.04	−0.12	0.04	0.08
≤1 bp for 3, >1.5 bp, persistent	140 (109)	−7.98*	−17.97**	−6.26	−15.19**	−0.21***	0.02	<0.01
≤1 bp for 3, >1.5 bp	229 (182)	−8.77*	−14.71**	−4.99	−15.20***	−0.45*	<0.01	<0.01
≤1 bp for 3, >2 bp, persistent	58 (41)	−4.37	−14.51	−1.53	−7.04	−0.22***	0.36	0.08
≤1 bp for 3, >2 bp	90 (67)	−6.64	−6.48	2.23	−5.21	−0.31*	0.11	0.29
≤1 bp for 3, >3 bp, persistent	19 (6)	4.92	−11.38	−0.68	−1.24	−0.31	0.66	0.26
≤1 bp for 3, >3 bp	27 (12)	0.83	−3.20	−0.82	−3.16	−0.27	0.93	0.80
≤1 bp for 5, >1.5 bp, persistent	77 (65)	−12.17**	−22.51***	−7.30	−23.65***	−0.10*	<0.01	<0.01
≤1 bp for 5, >1.5 bp	144 (123)	−14.93**	−23.68**	−6.82	−24.07***	−0.19*	<0.01	<0.01
≤1 bp for 5, >2 bp, persistent	26 (22)	−7.25	−24.92**	−8.08	−10.44	−0.12	0.23	<0.01
≤1 bp for 5, >2 bp	51 (44)	−15.48*	−23.84	−3.89	−13.40	−0.33	0.03	0.09
≤1 bp for 5, >3 bp, persistent	9 (6)	4.92	−11.38	−0.68	−1.24	−0.31	0.66	0.27
≤1 bp for 5, >3 bp	14 (10)	−0.51	−8.74	−2.82	−9.05	−0.29	0.96	0.47
From ≤0.5 to >3 bp in one step (Appendix A9)	24 (8)	−5.09	−25.43**	−11.13	1.94	−0.39	0.46	<0.01

Table A15: Entries of the main rule. The entries of the main rule of Table A13. Window a: the entry day lies in an auction window of the bond, from the announcement to the settlement of the auction. Window f: it lies in a futures delivery window, from the 20th of the month before a delivery month to the delivery day. Window y: it lies between 20 December and 5 January.

Bond	Entry day	Window	Bond	Entry day	Window
IT0004009673	18 Sep 2020		IT0005327306	22 Jan 2020	
IT0004604671	30 Dec 2019	y	IT0005329344	11 Nov 2020	
IT0004889033	18 Dec 2019		IT0005329344	30 Dec 2020	y
IT0004953417	31 Jan 2020		IT0005331878	18 Feb 2020	
IT0004953417	22 Dec 2020	y	IT0005344335	28 Oct 2019	
IT0005001547	20 Jul 2020		IT0005359846	26 Nov 2020	f
IT0005004426	21 Oct 2019		IT0005366007	28 Oct 2019	
IT0005056541	09 Nov 2020		IT0005367492	26 Jun 2020	
IT0005104473	22 Sep 2020		IT0005383309	22 Jan 2020	
IT0005104473	18 Dec 2020		IT0005399230	26 Aug 2020	a,f
IT0005246340	11 Aug 2020		IT0005399230	29 Sep 2020	
IT0005250946	28 Oct 2019		IT0005408502	26 Jan 2021	
IT0005321325	26 Jun 2020		IT0005422487	18 Nov 2020	

Columns (4) to (6) interact capped specialness with the two borrowing states. Panel C repeats Panel A with bond-by-month and day fixed effects and the life-cycle controls of Table A9. In the model without Assumption 1, a cost that bites only when a dealer must deliver more than it holds would reduce best-ask depth more for a special bond with above-average borrowing than for one with below-average borrowing. Best-ask depth falls in both special cells and not in the GC cell with above-average borrowing. Within bond-months (Panel C), the fall for special bonds with below-average borrowing is smaller and significant only at the 10% level, while the specialness slopes remain significant at the 1% level in both borrowing states. For best-ask depth, the difference between the two specialness slopes is not significant in any panel.

A13. Specialness and the repo bid-ask bounce

The MTS Repo records give the initiating side of each SC trade. Specialness measured from borrower-initiated trades exceeds specialness measured from lender-initiated trades by 0.51 bp on average, so the main measure moves with the mix of initiators. This gap differs from the gap of 0.58 bp in Appendix A10, which compares simple means of the repo rates within a bond-day. Each measure here instead weights trades by volume and compares each trade with the GC rate of its half hour. Table A17 re-estimates Table 3 of the paper with three alternatives. The first is the midpoint of the borrower-initiated and lender-initiated measures, on the 97% of the bond-days in the sample that have both. The second uses lender-initiated trades only. The third keeps the main measure and adds the borrower-initiated share of SC volume as a control. With bond and day fixed effects, the ratio, best-ask depth, best-bid depth and selling keep their signs and remain significant under all three. With bond-by-month and day fixed effects and the life-cycle controls of Table A9 (Panel B), the ratio and best-ask depth fall under every measure, whereas best-bid depth is not significant with lender-initiated trades only. With lender-initiated trades only, the estimates are about a quarter smaller with bond and day fixed effects and about a third smaller within bond-months.

Table A16: Special and borrowed. A bond-day has above-average borrowing if the log of the bond's borrower-initiated SC spot-next volume exceeds its mean over the previous 20 trading days (Panels A and C), or if its mean over the previous five trading days exceeds its mean over the 20 trading days before them (Panel B). Columns (1) to (3) report the difference from a bond at the GC rate, at most 0.5 bp, with below-average borrowing, for special bonds, above 2 bp, and bonds at the GC rate in the given borrowing state. The regressions also include an indicator for the bond-days between 0.5 and 2 bp. Columns (4) and (5) report the difference between a bond at the GC rate and a bond with 2 bp of specialness, twice the coefficient on capped specialness interacted with each borrowing state, with the borrowing indicator and the parts of specialness below zero and above 2 bp. Column (6) is the difference between columns (4) and (5). Panels A and B have bond and day fixed effects. Panel C has bond-by-month and day fixed effects and the life-cycle controls of Table A9. All regressions include the predetermined repo controls. Depth and the quoted spread are in log points and the executed sell in percentage points. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Cells, relative to GC with below-average borrowing			Capped specialness, GC rate to 2 bp		
	Special, above average (1)	Special, below average (2)	GC, above average (3)	Above-average borrowing (4)	Below-average borrowing (5)	Difference (6)
Panel A: Borrowing on the same day						
Ask/bid depth ratio	-14.74^{***}	-12.72^{***}	0.74	-17.27^{***}	-13.67^{***}	-3.60^{*}
	(2.23)	(2.27)	(1.71)	(2.53)	(2.39)	(1.86)
Best-ask depth	-8.87^{***}	-6.15^{***}	-1.54	-8.69^{***}	-7.02^{**}	-1.67
	(2.39)	(2.35)	(1.54)	(3.03)	(2.75)	(1.68)
Best-bid depth	5.87^{**}	6.57^{***}	-2.29	8.58^{***}	6.65^{**}	1.93
	(2.53)	(2.48)	(1.66)	(3.20)	(3.01)	(1.71)
Quoted spread	-1.30	-0.62	-2.11	1.56	0.59	0.97
	(2.16)	(2.09)	(1.37)	(2.68)	(2.56)	(1.62)
Any executed sell (pp)	-4.89^{**}	-7.35^{***}	1.43	-7.55^{***}	-9.61^{***}	2.06
	(1.97)	(1.93)	(1.45)	(1.88)	(1.98)	(1.45)
Panel B: Borrowing over the previous five days						
Ask/bid depth ratio	-13.77^{***}	-12.52^{***}	1.79	-17.59^{***}	-13.58^{***}	-4.01
	(2.49)	(2.48)	(2.18)	(2.50)	(2.73)	(2.42)
Best-ask depth	-7.88^{***}	-6.18^{**}	-0.92	-8.63^{***}	-7.35^{**}	-1.27
	(2.36)	(2.39)	(2.24)	(3.23)	(2.86)	(2.35)
Best-bid depth	5.89^{**}	6.34^{**}	-2.70	8.96^{***}	6.23^{**}	2.73
	(2.44)	(2.51)	(1.95)	(3.36)	(2.93)	(2.06)
Quoted spread	-0.15	0.36	-0.07	1.26	0.57	0.69
	(2.17)	(2.19)	(1.66)	(2.87)	(2.64)	(2.26)
Any executed sell (pp)	-5.25^{***}	-5.54^{***}	2.57^{*}	-9.25^{***}	-7.21^{***}	-2.05
	(1.78)	(1.85)	(1.51)	(1.99)	(2.02)	(1.71)
Panel C: Borrowing on the same day, bond × month and day FE and life-cycle controls						
Ask/bid depth ratio	-7.02^{***}	-5.99^{***}	-0.17	-8.69^{***}	-7.57^{***}	-1.11
	(1.83)	(1.85)	(1.54)	(1.78)	(2.02)	(1.52)
Best-ask depth	-4.51^{***}	-2.63^{*}	-0.78	-5.86^{***}	-4.48^{***}	-1.38
	(1.45)	(1.38)	(1.12)	(1.71)	(1.49)	(1.18)
Best-bid depth	2.52	3.36^{**}	-0.61	2.83	3.09^{*}	-0.26
	(1.62)	(1.59)	(1.19)	(1.84)	(1.80)	(1.18)
Quoted spread	-0.42	-0.52	-2.07	1.50	-0.51	2.00
	(1.84)	(1.81)	(1.25)	(1.95)	(1.91)	(1.49)
Any executed sell (pp)	1.04	-1.22	0.98	1.71	-0.41	2.13
	(2.10)	(2.03)	(1.62)	(2.03)	(2.23)	(1.54)

Table A17: Specialness and the repo bid-ask bounce. Each cell is the difference between a bond at the GC rate and a bond with 2 bp of specialness, twice the coefficient on capped specialness in equation (8) of the paper, with the predetermined repo controls. Panel A has bond and day fixed effects, and Panel B has bond-by-month and day fixed effects and the life-cycle controls of Table A9. Column (1) uses the main measure. Column (2) uses the midpoint of the measures from borrower-initiated and from lender-initiated SC trades, each computed as the main measure. Column (3) uses lender-initiated trades only. Column (4) uses the main measure and controls for the borrower-initiated share of SC volume. Depth and the quoted spread are in log points and the executed sell in percentage points. Standard errors clustered by bond and by day are in parentheses. *, ** and *** denote significance at the 10%, 5% and 1% levels.

	Main measure (1)	Midpoint of both initiators (2)	Lender-initiated trades only (3)	Main measure, share control (4)
Panel A: Bond and day FE				
Ask/bid depth ratio	-16.39^{***}	-15.67^{***}	-12.09^{***}	-16.16^{***}
	(2.28)	(2.25)	(1.95)	(2.24)
Best-ask depth	-7.99^{***}	-7.43^{***}	-6.04^{**}	-8.12^{***}
	(2.83)	(2.76)	(2.47)	(2.70)
Best-bid depth	8.41^{***}	8.24^{***}	6.05^{**}	8.04^{***}
	(3.09)	(3.05)	(2.85)	(2.97)
Quoted spread	1.35	1.21	−0.66	1.07
	(2.46)	(2.41)	(2.25)	(2.43)
Any executed sell (pp)	-8.73^{***}	-8.71^{***}	-6.90^{***}	-8.18^{***}
	(1.79)	(1.78)	(1.61)	(1.80)
Bond-days	36,622	35,634	35,773	36,622
Panel B: Bond × month and day FE and life-cycle controls				
Ask/bid depth ratio	-8.21^{***}	-7.68^{***}	-5.22^{***}	-8.27^{***}
	(1.68)	(1.65)	(1.34)	(1.69)
Best-ask depth	-4.66^{***}	-4.85^{***}	-3.08^{**}	-4.83^{***}
	(1.49)	(1.48)	(1.20)	(1.48)
Best-bid depth	3.55^{**}	2.83^{*}	2.14	3.45^{**}
	(1.70)	(1.68)	(1.41)	(1.69)
Quoted spread	0.88	0.86	−0.26	0.82
	(1.75)	(1.77)	(1.62)	(1.79)
Any executed sell (pp)	−0.02	0.03	−1.17	0.22
	(1.92)	(2.01)	(1.73)	(1.95)
Bond-days	36,622	35,634	35,773	36,622

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